

Robot Localization and Navigation

ROB-GY 6213

Lab 2 - Motion Modeling

Luca Macesanu, Kamil Imam, and Ethan Niv

Abstract—In order for mobile robots to be useful, they first need to understand where it is they are being used. Modern approaches to this problem - referred to as state estimation - tend to include a prediction and a correction step. What's more, when making predictions, it is useful to have quantifiable metrics for the uncertainty of these predictions, which can later be used when determining how to fuse predictions. In this work, we explore just the prediction component of this pipeline: profiling a car-like robot and creating an odometry model based on the bicycle motion model and using a single encoder.

I. INTRODUCTION

The development of a robust motion model for autonomous ground vehicles necessitates a balance between computational efficiency and predictive fidelity. While various kinematic representations exist, their efficacy is highly dependent on the vehicle's mechanical configuration and the environmental constraints. In this study, we implement a probabilistic bicycle model [1] to characterize the robot's trajectory. This modeling approach was selected over simpler unicycle [2] and differential drive [3] variants due to its superior ability to account for the physical geometry of the chassis and the specific distribution of the vehicle's mass. By integrating a probabilistic framework into the bicycle model architecture, we can more effectively compensate for the center of gravity and weight-loading effects, resulting in a more granular and accurate estimation of the robot's state-space transitions compared to standard linearized approximations [4].

II. METHOD

A. Modifications

For this lab we made minimal modifications to the system, only replacing a countersunk screw with a longer machine screw to prevent it from coming loose due to repeated use.

B. Bicycle Model Design

We explored various options for the motion model to implement for our car, and ended up between the bicycle model and differential drive model. Based on the physical implementation, we decided on the bicycle model, as the differential drive made unrealistic assumptions about the design of the robot, mainly unrealistic weight distribution, which the bicycle model accounts for. To implement the model, a few extra considerations needed to be taken, mainly the length of the wheelbase, and the radius from the rear tire to the center of rotation, with the wheelbase was measured



Fig. 1. **Robot Car:** Our robot car is equipped with rear wheel differential drive, front wheel steering, and a single rotary encoder for feedback control

to be 0.14m. The following equation is used, where L is the wheelbase, R is the radius, and α is the steering angle.

$$\tan(\alpha) = \frac{L}{R} \quad (1)$$

This is then redefined in terms of R , and the following equation $\Delta s = R \times \Delta\theta$ is subbed in for R , where Δs is the total distance traveled.

$$\Delta\theta = \frac{\Delta s}{L} \tan(\alpha) \quad (2)$$

This is further broken down into x and y components by first considering the robot is moving in a curve, and as such θ should be pulled from the middle point of the movement step.

$$\theta_{mid} = \theta_{current} + \frac{\Delta\theta}{2} \quad (3)$$

$$\begin{aligned} x_{new} &= x_{old} + \Delta s \cdot \cos(\theta_{mid}) \\ y_{new} &= y_{old} + \Delta s \cdot \sin(\theta_{mid}) \end{aligned} \quad (4)$$

III. EXPERIMENT DESIGN

We propose a number of experiments in order to characterize the behavior of the robot with a high degree of accuracy.



Fig. 2. **Testing environment:** In order to accurately measure error we create a high precision evaluation environment. At each meter of distance from the starting block we mark down the half centimeter, and use a high framerate camera (120fps) to measure at each distance.

A. Repeatable Testing Environment

In order to accurately model the behavior of the car we need an environment that is precise in its measurements and easy to set up. Our testing environment is made up of a 3 meter long strip of tape, with 4 lines intersecting it at 90 degrees at 0m, 1m, 2m, and 3m. Each intersecting line is marked to the nearest half centimeter, and there is a small tape arrow at the back of the robot that makes it easy to determine an accurate offset. For each experiment we follow the car with a high FPS camera recording the arrow passing over the distance markers, and log the error at each distance in a spreadsheet. At the start of the course there is a block placed that aligns the robot to have the same starting angle each time.

B. Steering Control

To get proper wheel alignment we first visually calibrate the steering angle. Next we collect 10 runs at that steering angle with the following procedure: Lift the robot off the ground, set the steering angle to -20, set the steering angle to the testing angle, place the robot against the starting block, start the recording, set the robot to max speed, and manually disable it once it has crossed the 3m mark. We follow these steps to minimize the variance of initial wheel angle that could be introduced by changing the angle while the robot is on the ground.

This procedure is then repeated until the mean error of two steering angles crosses the center line, at which point we know the optimal steering angle is between those two angles.

C. Mapping Encoder Counts to Distance

Wheel slip, damaged optical disks, cantilever drive shafts: there are many reasons that the distance calculated using an encoder may not reflect the ground truth. Following our steering alignment (to minimize the error in distance measurement from following curves) we attempt to profile the behavior of the encoder at lengths of time. We use the same setup configuration as the steering control experiment, but instead of manually stopping it after the 3m mark, we instead set it to run for varying times at max speed and measured the distance traveled.

D. Mapping Steering Commands to Rotational Velocity

In order to properly model the odometry of the vehicle, we must characterize the relationship between steering angle α_s and yaw rate $\dot{\theta}$. Visually estimating θ is difficult, so we instead rely on the fact that in a full rotation the vehicle will turn 360 degrees. As such we test steering angles $\alpha \in [10, 15, 20]$ in both directions, and measure the average lap time of the car 3 times. Dividing 360 by the average lap time will give us our estimated $\dot{\theta}$. While this approach will not give us ground truth angles every time, we find that it leads to more reliable estimates when compared with protractor use over shorter time domains.

E. Motion Model Real Trials

To evaluate the accuracy of the implemented bicycle model, we conducted a manual driving experiment. The robot was steered through a complex curve path while the system logged real-time encoder counts and steering commands. Upon completion of each trial, the physical end position (x_g, y_g) was measured using the tape-grid environment. This data was then processed to generate a predicted trajectory, which was compared against the physical ground truth to assess the model's fidelity in non-linear maneuvers.

F. Motion Model Sampling

To characterize the stochastic nature of the system, we implemented a sampling-based simulation framework. We executed 200 simulation iterations to model the propagation of uncertainty over a 5-meter trajectory. The simulation integrated the empirically derived variance functions, σ_s^2 and σ_ω^2 , to account for sensor noise and mechanical inconsistencies. A 2-Sigma boundary ellipse was calculated to encapsulate the distribution of final positions, allowing for a probabilistic validation of the bicycle model's reliability.

IV. EXPERIMENTAL RESULTS

In the following subsections we present the results of our experiments.

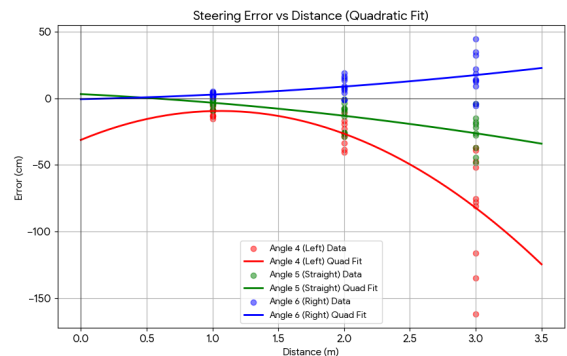


Fig. 3. **Dead Reckoning Error vs Steering Angle:** The following plot charts the the error in steering angle vs the location it traveled to. The error for 4 (Red) is large, while the error for 5 (Green) and 6 (Blue) is much closer, with 6 being better.

A. Steering Control

We initially visibly identified the optimal steering angle to be around 102 degrees on the servomotor, from where we ran tests at 3 steering angles, 4, 5 and 6; from where the ideal change in steering angle was adding 6 degrees, with an average variance of 2.575cm off the center line. We identified the optimal center angle as 108 degrees. Although the average error was relatively low, the standard deviation at 3 meters was quite high at 15.38cm. We believe that this is caused by the slack in the steering setup of the robot, as it has a decent amount of play, and as varies significantly when servo is turned. We also found that the ideal angle would be a middle value between 5 and 6 degrees, as 5 degrees rotated left and 6 rotated right.

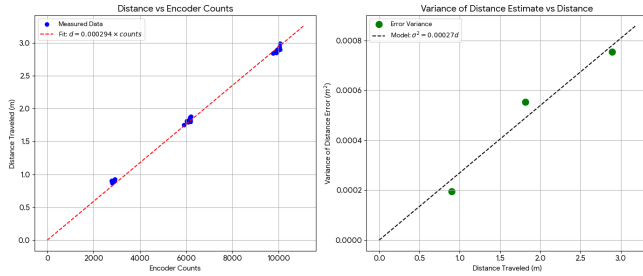


Fig. 4. **Encoder Counts vs Distance.** We find that variance increases as a function of the total distance travelled. This is inherent in state estimators that lack a correction step, and will be rectified in future work.

B. Mapping Encoder Counts to Distance

The encoder experimentation led us to get a mapping of 0.00294 meters per tick, with an average standard deviation at 3.21 cm per 80 ticks. We had a distance mapping that was incredibly accurate, likely thanks to the high fidelity of the encoder. Part of the gaps for standard deviation are explained by the fact that we calculated using absolute distance traveled in X and Y, rather than using the arcs that were taken by the robot in reality. There are three main sources of variance in the distance measurement: imperfections in the encoder (for instance, in an optical encoder if one of the windows is blocked), slippage from the wheels, and differences in the rotations between wheels. In the case of raw distance calculation, we believe slippage to be the main contributor to error on account of the dust that we observed accumulating on the wheels.

C. Mapping Steering Angle to Rotational Velocity

After experimental testing we found the bias of the system is evident with a yaw rate of -0.66 deg/s and slope of 2.25 degrees per second per change in steering command. While the linear relationship between steering angle and yaw rate is expected, the quadratic trend of the variance function is very surprising. During evaluation, the distance traveled increased as the steering command decreased in magnitude. Thus, we hypothesized that the greater distance would lead to a higher measured variance. The fact that the findings still indicate a quadratic increase in variance with steering angle magnitude

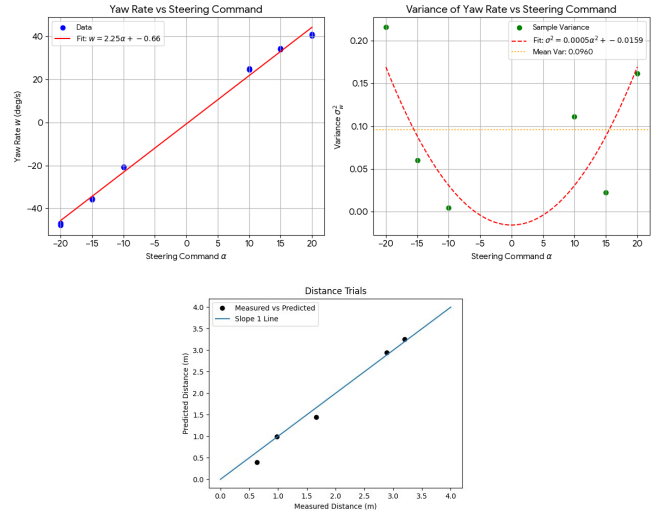


Fig. 5. **Yaw Rate vs Steering Command.** The bias of the system is evident and expected given experimental trials. Interestingly, the variance of the yaw rate follows a quadratic curve, contrary to our initial hypothesis about how steering angle affects model accuracy.

implies that the system dynamics are only stable for small turning angles. We theorize this may be due to increased slippage from the non-Ackerman front wheel configuration, whereby as the steering angle magnitude increases, the angle between wheels becomes progressively more acute.

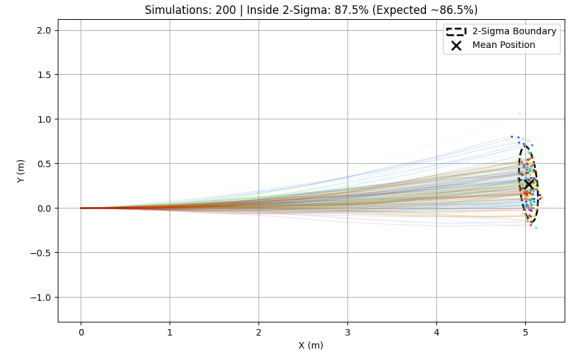


Fig. 6. **200 robot trial simulation results.** The motion model effectively represents the bias and variance inherent in our imperfect embodiment.

D. Motion Model Sim Trial

The experimental results demonstrate a high degree of precision in our deterministic model, with a mean testing error of less than 5 cm. Furthermore, the integration of noise parameters into the motion model allowed for a robust statistical representation of robot odometry. Monte Carlo simulations of the motion model yielded an 90.5 accuracy rate within the 2σ distribution when tested at a 4 degree steering angle, confirming our model is able to predict the movement accurately even when the incoming data has noise.

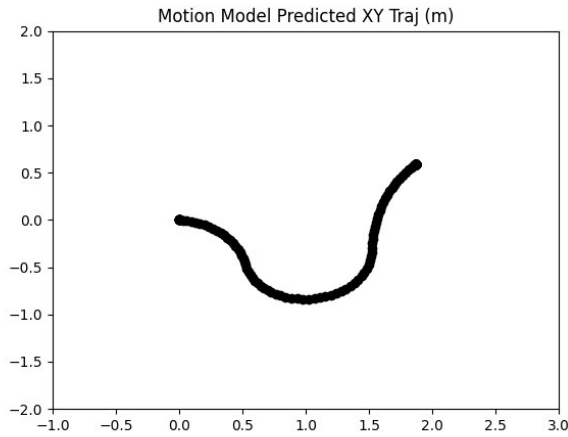


Fig. 7. **Complex Real Trial with Variable Steering Angle.** Error can be found in the overshoot calculated by the motion model, mainly introduced during the repeated sharp turns.

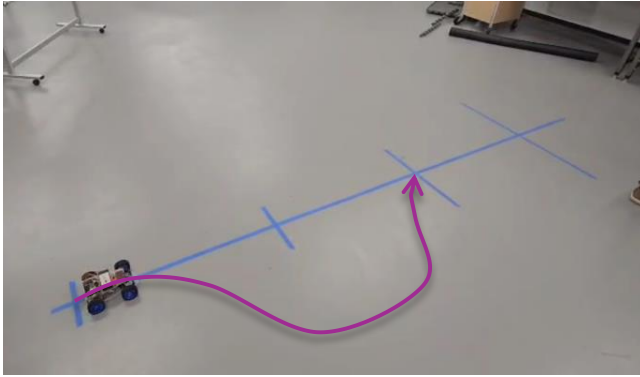


Fig. 8. **Complex Real Ground Truth Trajectory.** The trajectory was traced by overlaying the start and end frame of the video, and drawing a path along its motion

E. Complex Motion Model Real Trial

After testing the model with the data collected from a selection of our testing runs, we found that our model was within 5cm max of the absolute final location we measured physically, and the final test of the model using a user-controlled steering input closely matches the actual path of the car as seen in Figure 8 and the uploaded video. We attribute this close matching of the trajectory to the absolute trajectory to the accuracy of the bicycle model, as it is able to closely match the movement of the robot given only encoder and steering data.

V. CONCLUSION

This lab successfully established and validated the bicycle kinematic motion model through empirical characterization and stochastic analysis. Given the play in the steering mechanism, the drastic differences in the drive motors, suboptimal steering geometry, and slippage-prone wheels, our team is proud to have achieved such a degree of precision using only a single sensor to profile movement. While error still compounds quickly due to the lack of a correction step in

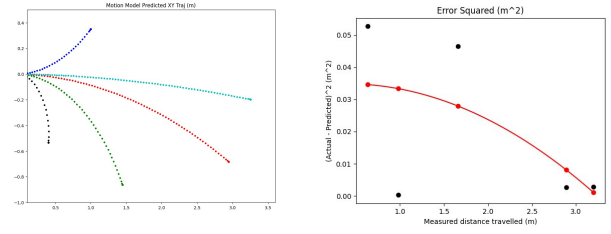


Fig. 9. **Basic Real Trials with Fixed Steering Angles and their error.** It is interesting to observe that the error decreases as the distance increases. This is because the longer distance measurements came from trials with lower magnitude steering angles, which, as a result of the imperfect steering geometry, lead to less slippage and thus lower error.

our state estimation, we are confident that future inclusion of this element will result in a stable and accurate system.

VI. FUTURE WORK

Our main goal is to replace the steering assembly to approximate an Ideal Ackerman drive [5]. This would allow us to drastically reduce front wheel slippage, thus making our odometry more reliable. In addition, further adjustments need to be made to the relative output velocities between the back two wheels in order to remove the bias in the current steering implementation. Finally we hope to build a suite of more complete analysis and testing tools, to allow for further profiling of the specific behaviors of the robot.

VII. REFERENCES

REFERENCES

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